

Andrew Charabin

Toronto, Ontario

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Work Experience

Loblaw Digital

Toronto, Ontario

Engineering Manager – Data Science & AI

Mar. 2025 – Present

- Recruited and led a team of 5 engineers accountable for machine learning & AI solutions spanning programmatic ad serving to campaign analytics at Advance, Loblaw's media platform.
- Launched an agentic service that creates and manages advertising campaigns exposed through a chat interface; drove a 10% sales lift through reduced friction in the sales funnel.
- Led development of a multi-agent service combining RAG-powered FAQ and analytics agents with persistent chat-history and context-aware authentication.
- Built an evaluation framework for a custom agentic workflow using recovery, tool success, and hallucination metrics; introduced LLM-as-a-judge evaluation and scaled a human-derived golden dataset; guiding refinements that achieved 100% task success rate.
- Increased sponsored search revenue by 15%+ by deploying a real-time click through rate (CTR) model to optimize ad placements.
- Designed a bid valuation service to recommend bid prices for ad opportunities; demonstrating a 20% increase in ROAS while maintaining media revenue.
- Launched an impression forecasting and booking service enabling a new display advertising product on Loblaw's web properties, generating \$15M in first-year revenue.

Greenstone Financial Services

Toronto, Ontario

Analytics Lead

May 2024 – Mar. 2025

- Led the Canadian analytics portfolio including business intelligence, marketing mix modelling, and agent lead allocation; recruited and managed a Data Analyst.
- Built a Bayesian marketing mix model to quantify channel-level incremental contribution, giving Sales and Marketing leadership a data-driven framework for budget allocation.

ADJ Industries

London, Ontario

Data Manager

Jun. 2023 – May 2024

- Enabled a business intelligence capability by implementing an on-premises MySQL database, saving \$50K+ annually in administration costs by centralizing data collection of payroll and customer-order data.

Voices

London, Ontario

Senior Manager, Data Science & Analytics

Oct. 2021 – Jun. 2023

- Instituted an organization-wide data science function by recruiting, onboarding, and mentoring 2 Data Scientists and 1 Data Analyst, creating a competitive edge for a global voice marketplace.
- Led a technology team responsible for centralized metrics, insights, and machine learning solutions across matching, pricing, and search to improve platform retention and liquidity.
- Reduced the time to market for new business intelligence reports and ML solutions by ~50% by leading the development of a Redshift data warehouse and ELT platform with modernized deployment, rollback, and monitoring.
- Increased job hire rate by 5 points by replacing rules-based matching with an XGBoost model that ranked voice talent by likelihood of hire.
- Contributed to a 3-point improvement in annual client retention by building a pricing recommendation system using SageMaker models to predict job price and response volume.

- Manager, Analytics Sep. 2020 - Oct. 2021
- Established a centralized analytics capability and common picture of company performance through Tableau core metrics and inferential and causal models, supporting product and marketing strategy during 30%+ YoY growth.
- Team Lead, Analytics Aug. 2019 - Sep. 2020
- Developed and standardized reporting and insights around platform liquidity metrics including churn, freelancer earnings, and spend retention, informing strategic discussion among leadership.
- Senior Financial Analyst Apr. 2018 – Jul. 2019
- Developed sales forecasts to guide financial targets and designed and rolled out sales comp. plans.

Skills

- AI & Engineering** Python, SQL, agentic AI, LLMs, RAG, evaluation
- Machine Learning** Regression, classification, recommender systems, ranking, forecasting
- Data Science** Experimentation, causal inference, marketing mix modelling
- Analytics** Product & marketing, metrics, insights, visualization, data storytelling
- Cloud** Google (Kubernetes, Argo CD, Vertex AI, BigTable), AWS (SageMaker, Lambda)
- Leadership** Team building, collaboration, structuring complexity, continuous improvement

Education

- Ivey School of Business London, Ontario
- Master of Science in Management (MSc), Business Analytics Jan. 2017 – Apr. 2018
- Honors Business Administration (HBA) Sep. 2013 – Apr. 2017
- Dean's Honor List 2014-2015 | Varsity Tennis Player, 2013 – 2018

Certifications

- LangChain Academy
- Certified Agent Engineer Oct. 2026
- University of Santa Cruz
- Bayesian Statistics Specialization Apr. 2024
- Python Institute
- Certified Associate Python Programmer Feb. 2024
- DeepLearning.AI
- Natural Language Processing Specialization Jan. 2024
- University of Pennsylvania
- A Crash Course in Causality: Inferring Causal Effects from Observational Data Aug. 2023
- AWS Training & Certification
- Certified Machine Learning Specialty Dec. 2022

Articles

- Goals, Principles & Constraints Mar. 2023
- A Framework for Organizational Leadership
- CI/CD for Multi-Model Endpoints in AWS Jun. 2023
- A simple, flexible alternative for sustainable ML solutions

Projects

Technical project portfolio available at acharabin.github.io.